Classical Field Theory — Exercise Sheet 3

Stress, improvements and angular momentum

Questions only

Use $T^{\mu}{}_{\nu} = p^{\mu}\partial_{\nu}\phi - \delta^{\mu}_{\nu}L$, $\eta = \mathrm{diag}(1,-1,\dots,-1)$ and the inverse-metric Hilbert convention of Lecture 3. All quantities below are smooth. The first problem has d spatial coordinates; the remaining problems have three. Problems 1–3 use Lectures 1–3. Problem 4 is a return problem after Lecture 9; its electromagnetic component conventions are supplied in the question.

Exercise 3.1 (Work by an external coupling). On $I \times \mathbb{T}^d$ consider

$$L = \frac{1}{2}\phi_t^2 - \frac{1}{2}|\nabla\phi|^2 - \frac{\lambda(t)}{4}\phi^4,$$

where I is an open interval and λ is prescribed and smooth.

(a) Derive the field equation and energy density. For a classical solution, find the derivative of its total energy, including the term due to explicit time dependence. Decide whether its physical spatial momentum $P^i = \int_{\mathbb{T}^d} T^{0i}\mathrm{d}^d x$ remains constant.

(b) Set $I = (0,\pi/2)$, $\lambda(t) = 1/\sin^2 t$ and $\phi(t,x) = \sin t$. Verify the field equation, compute its energy and check the sign and value of the energy derivative by both methods.

Exercise 3.2 (Angular momentum and weighted decay). For the autonomous scalar density $L = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - V(\phi)$ on $\mathbb{R}^{1+3}$, raise the second stress index with η. For fixed distinct spatial indices i, j, define

$$M^{\mu ij} = x^i T^{\mu j} - x^j T^{\mu i}.$$

(a) Prove its local conservation on solutions, keeping the derivatives of the explicit coordinate factors.

(b) Prove that $J^{ij}(t) = \int_{\mathbb{R}^3} M^{0ij}\mathrm{d}^3x$ is constant if, on every compact time interval, every component of $T^{\mu\nu}$ is bounded by $C(1+|x|)^{-4-\sigma}$ for some $C, \sigma > 0$. Explain why an unweighted integrability statement for stress alone does not automatically establish the angular charge hypotheses.

Exercise 3.3 (Belinfante improvement from a spin identity). Suppose a smooth tensor $T^{\mu\nu}$ on $\mathbb{R}^{1+3}$ is divergence-free, and a supplied smooth tensor $S^{\lambda\mu\nu}$ satisfies

$$S^{\lambda\mu\nu} = -S^{\lambda\nu\mu}, \qquad \partial_\lambda S^{\lambda\mu\nu} = T^{\nu\mu} - T^{\mu\nu}.$$

This spin identity is a given hypothesis; it is not inferred from scalar Lecture 3. Define

$$B^{\lambda\mu\nu} = \frac{1}{2}\big(S^{\lambda\mu\nu} + S^{\mu\nu\lambda} + S^{\nu\mu\lambda}\big), \qquad \Theta^{\mu\nu} = T^{\mu\nu} + \partial_\lambda B^{\lambda\mu\nu}.$$

Prove that Θ is symmetric and divergence-free. Whenever the slice integrals exist, find the difference between its integrated four-momentum and that of T. Give a sufficient boundary condition for equality, and explain why local improvement alone is insufficient.

Exercise 3.4 (Maxwell Hilbert stress, after Lecture 9). Return to this problem after Lecture 9. Let $F_{\mu\nu} = -F_{\nu\mu}$ be held fixed with both indices down while a Lorentz metric is varied. Use the density and Hilbert convention

$$L = -\frac{1}{4} g^{\mu\rho} g^{\nu\sigma} F_{\mu\nu} F_{\rho\sigma}, \qquad \delta S = \frac{1}{2} \int T_{\mu\nu} k^{\mu\nu} \mathrm{vol}_g, \quad k^{\mu\nu} = \delta g^{\mu\nu}.$$

The volume variation is $\delta \mathrm{vol}_g = -\frac{1}{2} g_{\mu\nu} k^{\mu\nu} \mathrm{vol}_g$. Derive $T_{\mu\nu}$ by varying both inverse metrics. At $g = \eta$, set $F_{i0} = E_i$ and $F_{ij} = \epsilon_{ijk} B_k$, with $\epsilon_{123} = 1$. Compute T_{00} and T^{i0} in terms of the vectors $\mathbf{E}$ and $\mathbf{B}$. Evaluate them for $\mathbf{E} = (1, 2, 0)$ and $\mathbf{B} = (0, 0, 3)$.